Plendl Hourglass Model - FRED edition, run 27 October 2026

Model version 1.0.0. Data: FRED (Federal Reserve Bank of St. Louis), downloaded 2026-10-27. Trend lines fitted over FY2000-FY2026. Projection starts 01 October 2026, monthly, 40 years.

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Headline

• Starting debt D0: $39.462 T (GFDEBTN, 2026-04-01)

• FY2026 receipts: $5.403 T; net interest: $1.044 T; primary deficit: $+0.948 T

• FY2026 interest-to-revenue (I/R): 19.3%; effective rate r0: 2.718%

• Currency in circulation: $2.416 T (2025-10-01), +2.6% year on year

• Observed NIPA interest / receipts (2026-04-01): 21.1% ($1.280 T / $6.056 T SAAR)

Calculated paradigm-shift point

Case	Paradigm-shift point	First month I/R >= 30%	First month I/R >= 40%
Case A: all trend lines fitted 2000-2026	None through September 2066	none	none
Case B: rate held at FY2026 level (dr = 0)	None through September 2066	Oct 2047	Apr 2059
Case C: rate slope fitted 2021-2026 only (+0.312 pp/yr)	June 2036 (I/R 55.5%)	Jun 2030	Mar 2033

Test: g_q(t) = 12 ln(q_t+1/q_t) and a(t) = g_q(t) - g_q(t-1). The paradigm-shift point is the first month from which, for every remaining month of the horizon, both g_q > 0 and a > 0. 30%/40% are reference lines only.

Fitted trend lines (OLS)

Series	Fit	Parameter	R^2
Receipts	log-linear	g_R = 4.21% / yr	0.917
Gross debt (reference)	log-linear	8.07% / yr	0.991
Effective rate (net interest / avg debt)	linear	dr = -0.0447 pp / yr	0.221
Primary deficit / receipts	linear	b_s = +1.221 pp / yr (start s0 = 17.6%)	0.174

Calculated values, yearly points (Case A; I/R for all cases)

Point	Debt $T	Interest $T	Revenue $T	Primary deficit $T	Rate %	I/R A %	I/R B %	I/R C %
Oct 2026	39.46	1.07	5.40	+0.95	2.72	19.8	19.8	19.8

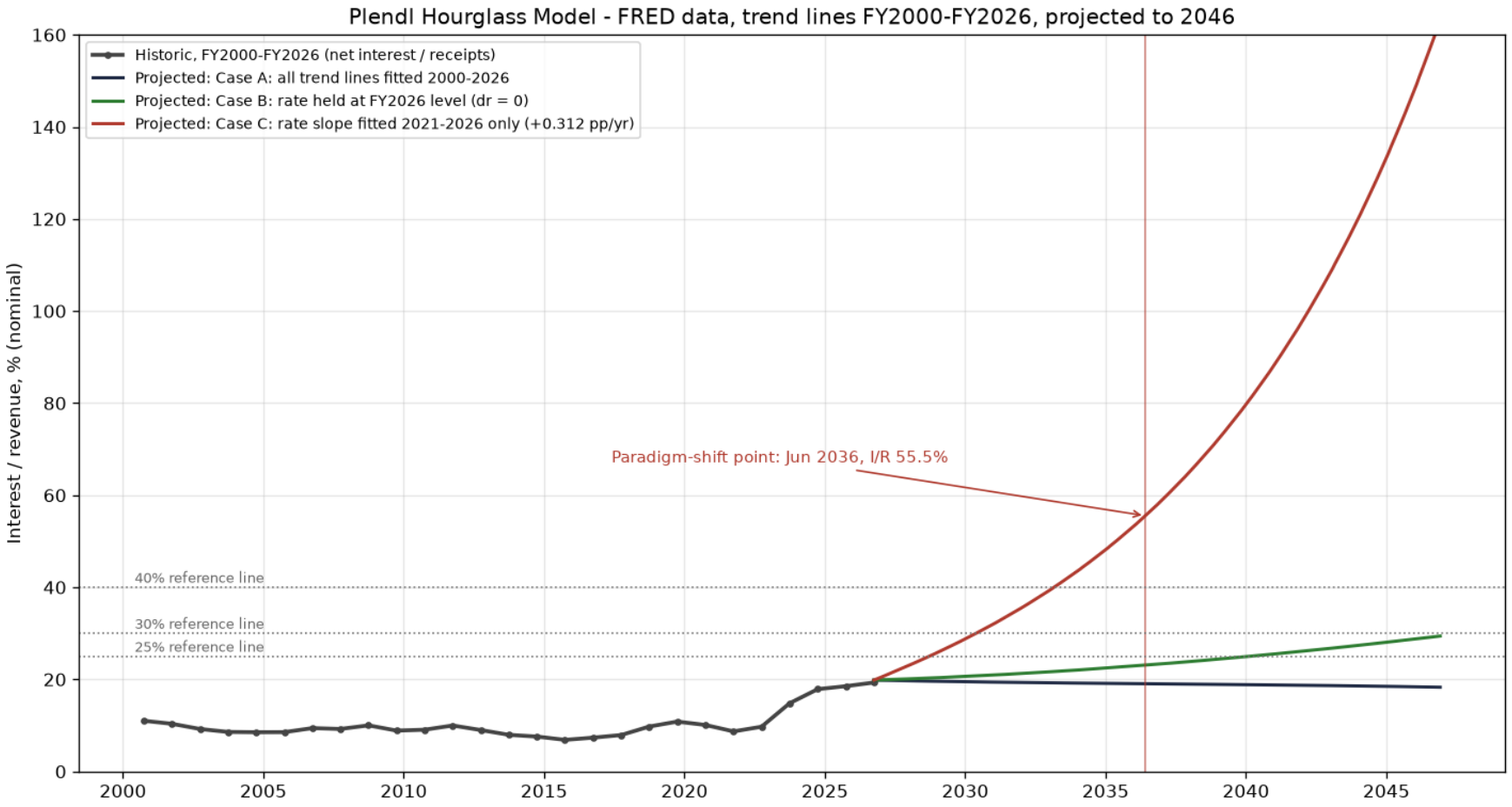
Point	Debt $T	Interest $T	Revenue $T	Primary deficit $T	Rate %	I/R A %	I/R B %	I/R C %
Oct 2027	41.55	1.11	5.63	+1.06	2.67	19.7	20.1	22.4
Oct 2028	43.79	1.15	5.87	+1.17	2.63	19.6	20.3	25.1
Oct 2029	46.19	1.19	6.11	+1.30	2.58	19.5	20.6	28.0
Oct 2030	48.76	1.24	6.37	+1.43	2.54	19.4	20.9	31.2
Oct 2031	51.51	1.28	6.64	+1.57	2.49	19.3	21.2	34.6
Oct 2032	54.46	1.33	6.92	+1.72	2.45	19.3	21.6	38.4
Oct 2033	57.61	1.39	7.21	+1.88	2.40	19.2	21.9	42.5
Oct 2034	60.98	1.44	7.51	+2.05	2.36	19.2	22.3	47.0
Oct 2035	64.58	1.49	7.83	+2.23	2.31	19.1	22.8	51.9
Oct 2036	68.42	1.55	8.16	+2.43	2.27	19.0	23.2	57.4
Oct 2037	72.52	1.61	8.50	+2.63	2.23	19.0	23.7	63.5
Oct 2038	76.90	1.68	8.86	+2.85	2.18	18.9	24.3	70.2
Oct 2039	81.57	1.74	9.23	+3.09	2.14	18.9	24.8	77.7
Oct 2040	86.54	1.81	9.62	+3.33	2.09	18.8	25.4	86.0
Oct 2041	91.83	1.88	10.02	+3.60	2.05	18.7	26.0	95.2
Oct 2042	97.47	1.95	10.45	+3.88	2.00	18.7	26.6	105.5
Oct 2043	103.46	2.02	10.89	+4.17	1.96	18.6	27.2	117.0
Oct 2044	109.83	2.10	11.34	+4.49	1.91	18.5	27.9	129.9
Oct 2045	116.61	2.18	11.82	+4.82	1.87	18.4	28.6	144.2
Oct 2046	123.80	2.26	12.32	+5.17	1.82	18.3	29.3	160.3

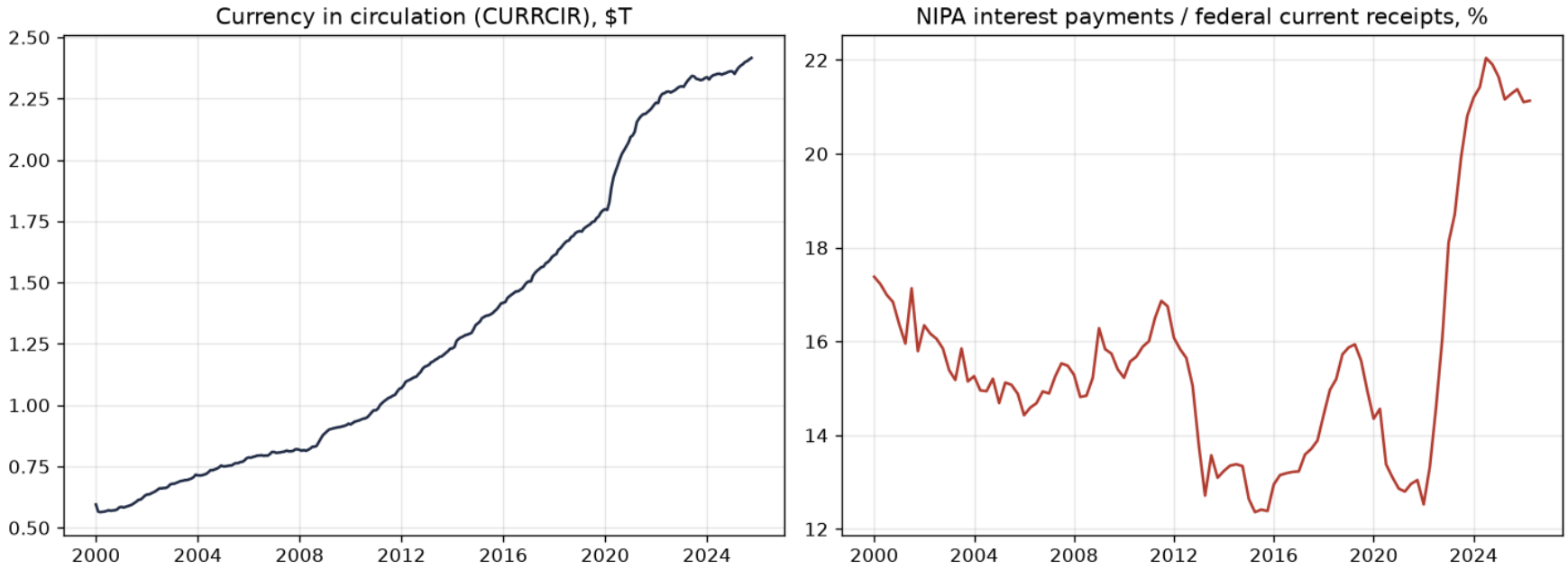
Historic data (FRED)

FY	Receipts $T	Outlays $T	Net interest $T	Primary deficit $T	Gross debt $T	Eff. rate %	I/R %
2000	2.025	1.789	0.223	-0.459	5.63	3.969	11.0
2001	1.991	1.863	0.206	-0.334	5.77	3.617	10.4
2002	1.853	2.011	0.171	-0.013	6.20	2.857	9.2
2003	1.782	2.160	0.153	+0.225	6.76	2.363	8.6
2004	1.880	2.293	0.160	+0.252	7.35	2.271	8.5
2005	2.154	2.472	0.184	+0.134	7.91	2.411	8.5
2006	2.407	2.655	0.227	+0.022	8.45	2.771	9.4
2007	2.568	2.729	0.237	-0.076	8.95	2.725	9.2
2008	2.524	2.983	0.253	+0.206	9.99	2.669	10.0
2009	2.105	3.518	0.187	+1.226	11.88	1.710	8.9
2010	2.163	3.457	0.196	+1.098	13.53	1.545	9.1
2011	2.303	3.603	0.230	+1.070	14.76	1.626	10.0
2012	2.450	3.527	0.220	+0.856	16.05	1.431	9.0

FY	Receipts $T	Outlays $T	Net interest $T	Primary deficit $T	Gross debt $T	Eff. rate %	I/R %
2013	2.775	3.455	0.221	+0.459	16.72	1.348	8.0
2014	3.021	3.506	0.229	+0.256	17.79	1.327	7.6
2015	3.250	3.692	0.223	+0.219	18.12	1.243	6.9
2016	3.268	3.853	0.240	+0.345	19.54	1.275	7.3
2017	3.316	3.982	0.263	+0.403	20.21	1.321	7.9
2018	3.330	4.109	0.325	+0.454	21.46	1.560	9.8
2019	3.463	4.447	0.375	+0.608	22.67	1.700	10.8
2020	3.421	6.517	0.345	+2.750	26.90	1.394	10.1
2021	4.047	6.821	0.352	+2.421	28.39	1.275	8.7
2022	4.897	6.272	0.476	+0.898	30.84	1.607	9.7
2023	4.441	6.128	0.658	+1.029	32.99	2.063	14.8
2024	4.920	6.735	0.880	+0.935	35.23	2.580	17.9
2025	5.236	7.011	0.970	+0.805	37.38	2.672	18.5
2026p	5.403	7.396	1.044	+0.948	39.46	2.718	19.3

Note: FY2026 is preliminary: receipts/outlays = sum of MTS Oct 2025-Sep 2026; net interest = FY2025 x 1.0763 (NIPA interest growth, last 4 vs prior 4 quarters, through 2026-04-01); debt = GFDEBTN 2026-04-01.


Plendl Hourglass Model - FRED data, trend lines FY2000-FY2026, projected to 2046
Historic, FY2000-FY2026 (net interest / receipts)
Projected: Case A: all trend lines fitted 2000-2026
Projected: Case B: rate held at FY2026 level (dr = 0)
Projected: Case C: rate slope fitted 2021-2026 only (+0.312 pp/yr)
Interest / revenue, % (nominal)
Paradigm-shift point: Jun 2036, I/R 55.5%
40% reference line
30% reference line
25% reference line
0
20
40
60
80
100
120
140
160
2000
2005
2010
2015
2020
2025
2030
2035
2040
2045



FRED series

- FYFR: Federal Receipts, fiscal year (millions $, annual FY) - https://fred.stlouisfed.org/series/FYFR
- FYONET: Federal Net Outlays, fiscal year (millions $, annual FY) - https://fred.stlouisfed.org/series/FYONET
- FYOINT: Federal Outlays: Interest (net interest), fiscal year (millions $, annual FY) - https://fred.stlouisfed.org/series/FYOINT
- FYGFD: Gross Federal Debt, fiscal year end (billions $, annual FY) - https://fred.stlouisfed.org/series/FYGFD
- GFDEBTN: Federal Debt: Total Public Debt (millions $, quarterly) - https://fred.stlouisfed.org/series/GFDEBTN
- MTSR133FMS: Monthly Treasury Statement: Total Receipts (millions $, monthly) - https://fred.stlouisfed.org/series/MTSR133FMS
- MTSO133FMS: Monthly Treasury Statement: Total Outlays (millions $, monthly) - https://fred.stlouisfed.org/series/MTSO133FMS
- A091RC1Q027SBEA: Federal government current expenditures: Interest payments (billions $ SAAR, quarterly) - https://fred.stlouisfed.org/series/A091RC1Q027SBEA
- FGRECPT: Federal government current receipts (billions $ SAAR, quarterly) - https://fred.stlouisfed.org/series/FGRECPT
- CURRCIR: Currency in Circulation (billions $, monthly) - https://fred.stlouisfed.org/series/CURRCIR